Gina & 宏孝的SDGs多元選修課

From Waste to Worth

Circular Economy & Climate Change · Student Worksheet

Name:	Class:	No.:	Group:

Today's Mission — By the end of this lesson, I can:
☐ explain the difference between a **linear** and a **circular** economy
☐ explain **why** a circular economy can slow down climate change
☐ describe and judge what six well-known companies are doing
☐ design a simple circular solution for something at our school

Key Words

Word	Meaning (in simple English)
linear economy	take → make → use → throw away
circular economy	keep products and materials in use for as long as possible
raw materials	things we take from nature to make products (metal, wood, cotton, oil)
greenhouse gas emissions	gases like CO_2 that trap heat and warm the planet
carbon footprint	the total greenhouse gas created by a product or a person
landfill	a place where waste is buried in the ground
durable	strong; able to last a long time
repair / resell / recycle	fix it / sell it again / turn it into new material
greenwashing	making a company look greener than it really is

Task 1 · Warm-up: Circular Economy Around Us (5 min, pairs)

A. Match each everyday example to one circular idea: share / reduce / reuse / repair / recycle

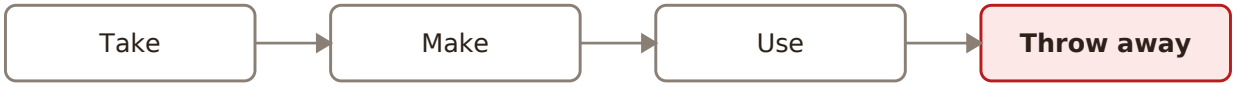
Everyday example	Circular idea
YouBike: one bike is ridden by many different people every day.	
Bringing your own cup to a drink shop.	
Families sort plastic, paper and cans for the recycling truck.	
Dropping old batteries in the box at a convenience store.	
Older students pass down uniforms and textbooks to younger students.	
Getting your shoes or umbrella fixed at a market repair stall.	

B. Think. Why is one YouBike shared by many people better for the climate than everyone buying their own bike?

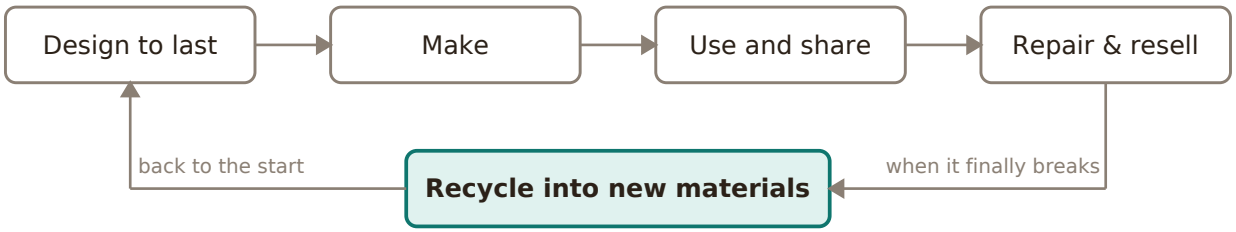
C. My life. Which of these did I use this week? ☐ YouBike ☐ own cup ☐ recycling ☐ battery box ☐ hand-me-downs ☐ repair
One more circular example I see in my life:

Task 2 · Linear vs Circular: Where Is the Carbon? (8 min)

Linear economy



Circular economy



A. Look at the diagram and fill in the blanks.

In a linear economy, we _____, _____, use, and then _____ things. In a circular economy, we design products to _____, we _____ and resell them, and at the end we _____ them into new materials.

B. Fact box: the missing half

The Ellen MacArthur Foundation says that about **55%** of global greenhouse gas emissions come from **energy** (electricity, heating, transport). The other **45%** come from **making things** we use every day, like cars, clothes and food, and from how we use land. So clean energy alone can only fix about half of the climate problem.

C. Complete the cause-and-effect chain.

We buy a new phone every 2 years → more _____ must be dug out of the ground → factories use more _____ → more _____ go into the air → the planet gets _____.

Now change one link: *If we keep our phone for 4 years instead of 2, …*

D. Think deeper: Which "R" helps the climate most? Rank them 1 (most helpful) to 4 (least helpful).

Strategy	Rank
Reduce: buy fewer new things	
Reuse: give or sell it to someone else	
Repair: fix it so it lasts longer	
Recycle: break it down into new materials	

One reason for my No. 1:

Task 3 · Jigsaw: Company Detectives (18 min)

Step 1 · Read (8 min). Your teacher gives you two cards: **A + D**, **B + E** or **C + F**. Read your two cards and fill in those two rows of My Detective Notes. You may check your answers with students from other groups who have the same cards. (In a group of four, two students share one pair of cards.)

Step 2 · Teach (8 min). Go back to your home group. Teach your two companies in 2 minutes. Listen to your group members and fill in the other four rows.

Step 3 · Report (2 min). Your teacher will pick one group to share.

Card A · Apple: Making New Phones from Old Ones

Apple wants to one day make all its products from only recycled and renewable materials. In 2025, 30% of all the materials in the products it shipped came from recycled sources, the highest level in the company's history. Every battery that Apple designs now uses 100% recycled cobalt, and all its magnets use 100% recycled rare earth elements. Apple also removed plastic from its packaging. To get materials back, Apple built a robot called **Daisy** that takes old iPhones apart, so recyclers can recover metals like cobalt and lithium. Apple's goal is to make every product carbon neutral by 2030. However, Apple still launches new iPhone models every year.

Card B · IKEA: Buying Back Your Old Furniture

IKEA is the world's largest furniture chain. Its furniture is cheap and easy to carry home, so many people buy it and then throw it away when they move. To change this, IKEA started a **Buy Back & Resell** service in many countries. Customers bring back used IKEA furniture, such as bookcases, desks, small tables and chairs, if it is still in good condition. IKEA checks it and pays the customer in store credit. Then the furniture is sold again at a lower price in a special area of the store. Items that cannot be sold again may be recycled or donated. Beds, sofas and mattresses usually cannot be returned. IKEA's goal is to become a circular and “climate positive” business by 2030.

Card C · Patagonia: Fix It, Don't Buy It

Patagonia is an outdoor clothing company. In 2013 it started a programme called **Worn Wear**. The message is simple: take care of your clothes, repair them, and buy used when you can. Customers can trade in old Patagonia clothes, and the company sells them again second-hand. Patagonia also repairs clothes, often for free. In Europe alone, it repaired more than 35,000 pieces of clothing in 2025, and it plans to repair 100,000 pieces a year by 2028. A Worn Wear repair truck travels to ski towns, fixes jackets from any brand, and teaches people to fix their own gear. The repair team also tells designers what breaks most often, so new products can last longer.

Card D · Nike: Old Shoes, New Playgrounds

Nike is one of the world's biggest sportswear companies. Since 1992, a programme called **Nike Grind** has turned factory waste and worn-out shoes into new materials. These materials are used to build running tracks, gym floors, playgrounds and basketball courts. So far, Nike Grind has recycled more than 155 million pounds (about 70 million kg) of waste. Nike also runs **Nike Refurbished**: shoes that customers return in like-new condition are checked, cleaned and sold again at a lower price. More than two million products have been sold this way. People can also bring old shoes and clothes to some Nike stores, where they are donated or recycled. However, Nike still sells millions of new pairs of shoes every year.

Card E · Uniqlo: Repair, Remake, Reuse, Recycle

Uniqlo is a Japanese clothing company with stores all over Asia, including Taiwan. Its programme **RE.UNIQLO** has four parts. **Repair:** in all its UK stores, staff can fix holes and broken parts of your clothes, starting from just £2. **Remake:** in special RE.UNIQLO Studios, you can add embroidery or Japanese *sashiko* patches to give old clothes a new look. There are 21 of these studios across Europe. **Reuse:** Uniqlo collects clothes that people no longer wear and donates them to people who need them. **Recycle:** clothes that cannot be worn any more are turned into raw materials for new clothes or into energy. However, Uniqlo's clothes are cheap, so many people buy a lot of them.

Card F · Coca-Cola: The Plastic Bottle Problem

Coca-Cola sells drinks all around the world, and many of them come in single-use plastic bottles. In 2018, it started a plan called **World Without Waste**. It promised that by 2030, at least 50% of the material in its packaging would be recycled. In 2022, it added another promise: by 2030, 25% of its drinks would be sold in refillable or returnable bottles, or from fountain machines. But in December 2024, Coca-Cola changed its goals. The new goal is 35–40% recycled material by 2035, and the promise about refillable bottles was removed. Coca-Cola says the old goals were very difficult to reach. Some environmental groups called the change “a masterclass in greenwashing.”

My Detective Notes

Company	What do they do?	Which “R”? (reduce / reuse / repair / recycle)	How does this cut emissions?	One question or doubt I have
A · Apple				
B · IKEA				
C · Patagonia				
D · Nike				
E · Uniqlo				
F · Coca-Cola				

Useful language for teaching your group

• My company is _____. The main idea is that _____. • For example, _____.

• This helps the climate because _____. • I'm not sure if _____, because _____.

Task 4 · Think Deeper: How Circular Is It Really? (10 min)

A. Circular Score (home groups). Give each of the six companies 1 (not at all) to 5 (very much) for each question.

Question	Apple	IKEA	Pata-gonia	Nike	Uniqlo	Coca-Cola
1. Does it keep products in use for longer?						
2. Does it reduce the need for new raw materials?						
3. Does it help people buy **less**, not just buy "greener"?						
Total (out of 15)						

Our highest score went to _____ because

B. Greenwashing detector. These are made-up company slogans. Write **R** (real circular action) or **G** (possible greenwashing). Explain in a few words.

Slogan	R or G?	Why?
"Our new T-shirt has 20% recycled plastic, so buy one in every colour!"		
"Bring back your broken shoes. We will fix them for free."		
"Our bags are 100% natural and eco-friendly!" (no details)		
"Return your old laptop and we will reuse its parts in new ones."		

C. Take a stand. Move to the corner that matches your opinion:

"A company that sells a new model every year can never be truly circular."
Strongly agree / Agree / Disagree / Strongly disagree

My opinion: I _____ because

My evidence from a case card:

Someone might say _____, but I think

Task 5 · Circular School Challenge + Exit Ticket (10 min)

A. Design challenge (home groups, 5 min). Choose ONE thing at our school that often becomes waste. Design a circular solution like a company would.

Circle one: school uniforms / PE shoes / old textbooks / plastic lunch boxes / broken umbrellas / my own idea: _____

Question	Our plan
Name of our project	
What is the problem now? (Where does it end up?)	
Our idea: which “R” do we use?	
How will it work? (who, where, when)	
How does it cut greenhouse gas emissions?	
One challenge, and how we solve it	

30-second pitch: “Our project is called _____. Right now, _____ ends up in _____. With our idea, students can _____. This helps the climate because _____.”

B. Exit ticket 3-2-1 (on your own, 3 min)

3 words I learned today: _____ / _____ / _____

2 ways a circular economy helps slow climate change:

1 thing I will change in my own life, starting this week:

C. Self-check — tick what you can do now.

☐ I can explain the difference between a linear and a circular economy.
☐ I can explain why making things creates greenhouse gas.
☐ I can describe one company's circular action and judge it.
☐ I can use evidence to support my opinion in English.